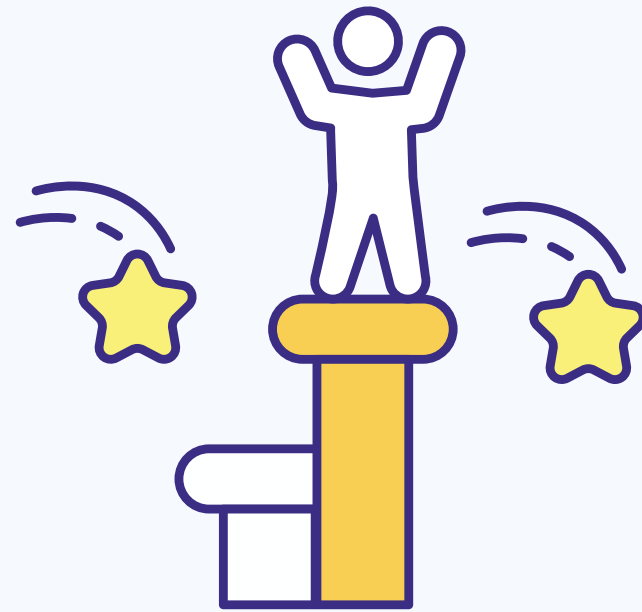




DEVOPS FUNDAMENTALS

Novice Level



1 - DevOps Pre-Requisites

2 - DevOps Fundamentals

3 - DevOps Core

4 - DevOps Advanced



TECHWORLD
WITH NANA



2 - DevOps Fundamentals

Week 4

Weekly Progress Tracker

START Cloud - 5.01: Introduction to Cloud & IaaS
END Nexus - 6.05: Publish Artifact to Repository

Key Topics for the Week

- **Cloud and IaaS Concepts:** High-level overview of cloud computing, focusing on Infrastructure as a Service (IaaS) and its key components
- **Deploying Applications:** How to build a JAR file and run an application on cloud infrastructure
- **Repository Management:** Learn about repository managers, like Nexus, their purpose, and key features
- **Nexus Installation and Configuration:** How to install Nexus on a DigitalOcean droplet, configure its service, and access it via a browser
- **Managing Repositories:** How to create a proxy repository in Nexus, configure roles, and publish an application by updating the build.gradle file

Checklist for This Week

- ☐ Create and configure a DigitalOcean droplet with the correct firewall settings and Java installed
- ☐ Build a JAR file locally and deploy it to the droplet
- ☐ Add a new user to the droplet and configure user permissions
- ☐ Install and configure Nexus
- ☐ Set up a Docker proxy repository and publish a build by updating your build.gradle file

Recommendations & Tips

- **Remember:** Make sure to configure your droplet with enough resources to run Nexus without any performance issues

Week 5

Weekly Progress Tracker

START Nexus - 06.06: Nexus REST API
END Docker - 07.07: Docker Demo - Project Overview

Key Topics for the Week

- **Nexus Repository Manager:** Understand GET request responses from a Nexus server, blob stores, and storage considerations
- **Repository and Cleanup Policies:** Deep dive into creating repositories, setting cleanup policies, and scheduling tasks to manage storage efficiently
- **Containers and Docker Basics:** Introduction to containers and Docker, covering installation and essential commands such as - *docker run, docker pull, docker ps, docker stop, docker logs, and docker exec*
- **Docker vs VMs:** How Docker differs from virtual machines
- **Docker Architecture:** Docker Server, Docker API, Docker CLI

Checklist for This Week

- ☐ Demo a GET request from the Nexus server and verify the response
- ☐ Create a blob store in Nexus and configure storage
- ☐ Set up a repository and define a cleanup policy in Nexus
- ☐ Install Docker and run basic commands (*docker run, docker pull, docker ps*)
- ☐ Run more advanced Docker commands like **docker stop** and **docker logs** to debug containers

Recommendations & Tips

- **Tip:** The final lecture this week will give you an overview of the upcoming demos that follow next week, ensure you take this information on board before starting work again next week

Week 6

Weekly Progress Tracker

START Docker - 07.08: Developing with Containers
END Docker - 07.13: Docker Volumes - Persisting Data

Key Topics for the Week

- **Docker Networking:** How Docker networks allow containerized services to communicate
- **Docker Compose:** Understand how Docker Compose can be used to build and deploy an application
- **Dockerfile & Image Building:** Learn how to write a Dockerfile and use it to build a custom image
- **AWS ECR:** Learn how to push and pull images from Amazon Elastic Container Registry (ECR) and manage Docker image tagging
- **Volumes in Docker:** Understand how Docker volumes manage persistent storage for containers

Checklist for This Week

- ☐ Set up Mongo Express with Docker and create a Docker network for communication between containers
- ☐ Use Docker Compose to deploy a multi-container application
- ☐ Write a Dockerfile and build a custom image
- ☐ Push and pull Docker images to/from AWS ECR, including image tagging
- ☐ Implement Docker volumes to manage persistent data in your containers

Recommendations & Tips

- **Reminder:** When logging in to the Mongo Express UI, the default username is **admin** and **pass**



2 - DevOps Fundamentals

Week 7

Weekly Progress Tracker

START Docker - 07.14: Docker Volumes Demo

END Docker - 07.17: Docker Best Practices

Key Topics for the Week

- **Adding Volumes to Docker Applications:** Learn how to attach Docker volumes to a Node.js application for persistent data storage
- **Nexus for Docker Image Hosting:** Understand how Nexus can be used as a private Docker registry to host Docker images
- **Running Nexus in Docker:** Set up and run Nexus as a Docker container using the official Nexus Docker image
- **Best Practices for Docker:** Explore essential best practices for managing Docker images, containers, and volumes

Checklist for This Week

- ☐ Add Docker volumes to your Node.js application for data persistence
- ☐ Configure Nexus to host Docker images and manage a private Docker registry
- ☐ Run Nexus as a Docker container using the official Nexus image

Recommendations & Tips

- **Reminder:** Nexus can be used to store multiple types of artifacts, not just Docker images